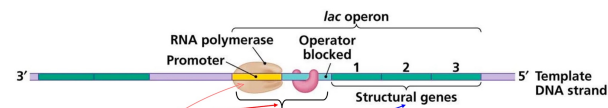


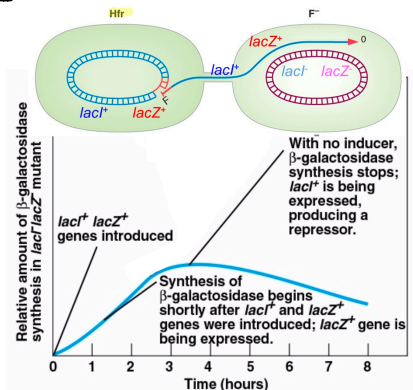
chap1 基因表达调控和表观遗传分析

一. 原核生物的基因表达调控 —— 乳糖操纵子



- 两个调控位点（顺式作用元件）：
 (1) 启动子 *lacP*：RNA聚合酶的结合位点，
 (2) “操纵基因” *lacO*：阻遏物结合位点。
- 三个结构基因：
 (1) *lacZ*：编码β-半乳糖苷酶，
 (2) *lacY*：编码β-半乳糖苷透性酶，
 (3) *lacA*：编码疏基半乳糖苷转乙酰酶。
 构成一个多顺反子转录单位，由一个启动子控制。

1. PaJamO 实验



证实了 *lacI* 编码 repressor

2. 表型分析

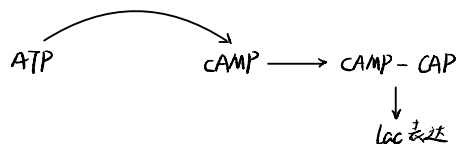
Genotype	β-galactosidase (<i>lacZ</i>)		permease (<i>lacY</i>)		Conclusions
	- IPTG	+ IPTG	- IPTG	+ IPTG	
<i>I⁺ Z⁺ Y⁺</i>	-	+	-	+	Wild type is inducible.
<i>I⁺ Z⁺ Y⁻</i>	+	+	+	+	<i>lacI</i> is constitutive.
<i>I⁺ Z⁺ Y⁻ / F' I⁺ Z⁺ Y⁺</i>	-	+	-	+	<i>lacI</i> is dominant to <i>lacI</i> .
<i>I⁺ Z⁺ Y⁻ / F' I⁺ Z⁺ Y⁻</i>	-	+	-	+	<i>lacI</i> is trans or cis-acting?

Genotype	β-galactosidase (<i>lacZ</i>)		permease (<i>lacY</i>)		Conclusions
	- IPTG	+ IPTG	- IPTG	+ IPTG	
<i>I⁺ Z⁺ Y⁺</i>	-	+	-	+	Wild type is inducible.
<i>I⁺ Z⁺ Y⁺</i>	-	-	-	-	<i>lacI^S</i> is uninducible.
<i>I⁺ Z⁺ Y⁺ / F' I⁺ Z⁺ Y⁺</i>	-	-	-	-	<i>lacI^S</i> is Q or R to <i>lacI⁺</i> .

Genotype	β-galactosidase (<i>lacZ</i>)		permease (<i>lacY</i>)		Conclusions
	- IPTG	+ IPTG	- IPTG	+ IPTG	
<i>O⁺ Z⁺ Y⁺</i>	-	+	-	+	Wild type is inducible.
<i>O⁺ Z⁺ Y⁺ / F' O⁺ Z⁺ Y⁺</i>	-	+	-	+	<i>Z⁺</i> is dominant to <i>Z⁻</i> .
<i>O^C Z⁺ Y⁺</i>	+	+	+	+	<i>O^C</i> is constitutive.
<i>O⁺ Z⁺ Y⁺ / F' O^C Z⁺ Y⁺</i>	+	+	-	+	<i>O^C</i> is cis-acting
<i>I⁺ Z⁺ Y⁺ / F' I⁺ Z⁺ Y⁻</i>	-	+	-	+	<i>lacI</i> is trans-acting

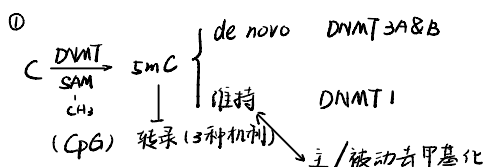
3. CAP / CRP

Glc ↑ → 腺苷酸环化酶

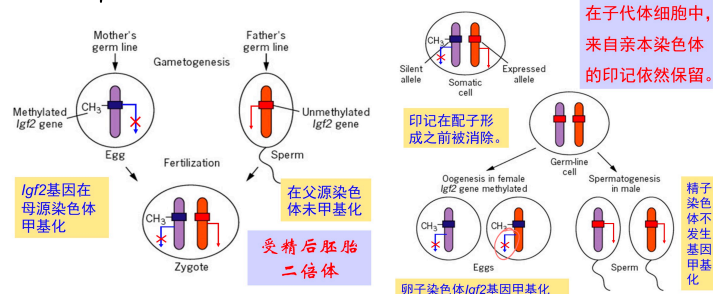


二. 真核生物基因表达调控和表观遗传分析

1. DNA 甲基化

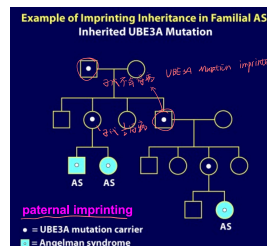


② 基因印记 (IGF-2)

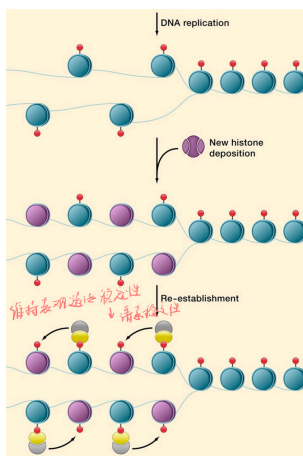


PWS, AS 综合征

maternal imprinting
 paternal imprinting



2. 组蛋白修饰 / 变体



H3K9me → 异染色质

DNMT1 ↔ HDAC

H2A.Z → 表达

H2A.W → 表达

3. 染色质重塑

耗能, CRC